

The Role of AI

Algorithm to bedside symposia





DATA STORIES

From algorithm to bedside

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Senior Forsker

Gentofte or Herlev Hospital

03-14 Feb. 2025

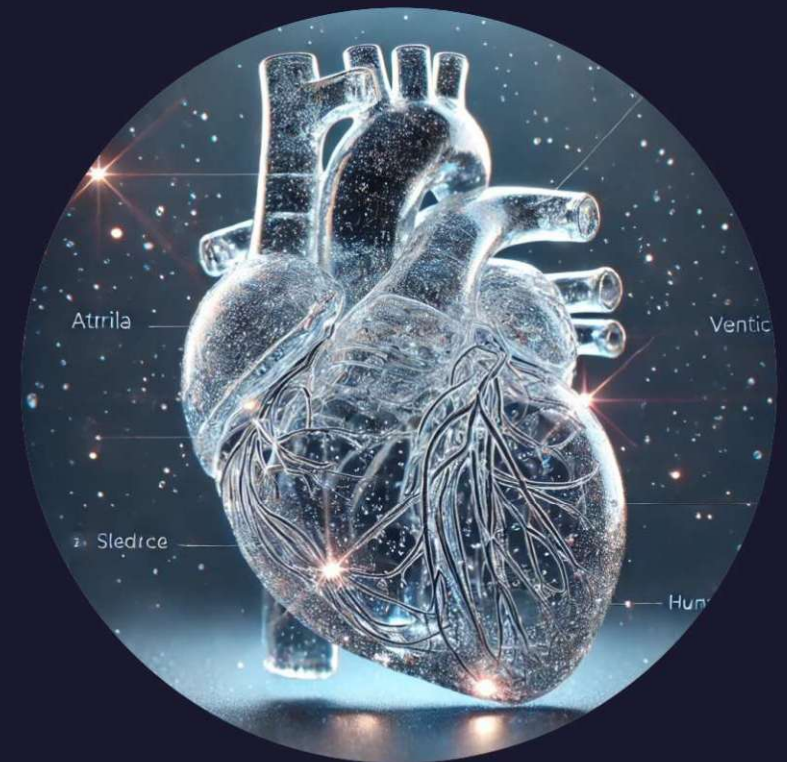
Lunch symposia

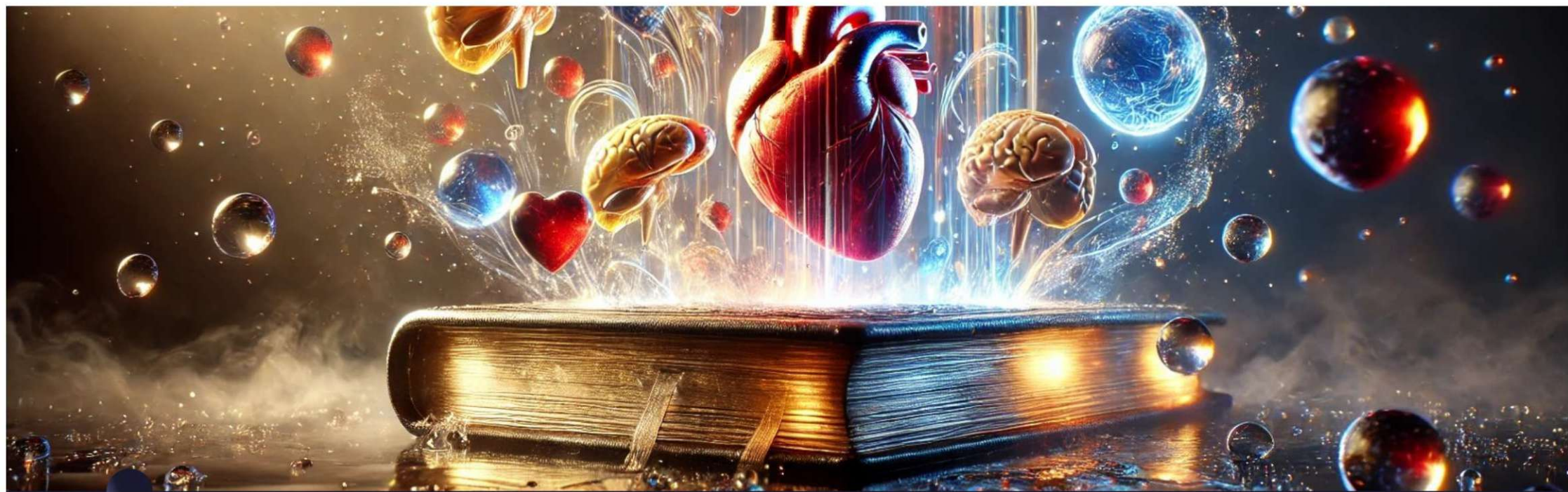
Omics-Based Clinical and Population Studies (**session 1: Mon. 3 Feb. 2025** - Specific Case Studies)

Contextual Epidemiology of Cardiometabolic Conditions, Polypharmacy and Multi-morbidity (**session 2: Thurs. 6 Feb. 2025**)

Scientific Methods in Biomedical Research (**sessions 3 and 4: Mon and Thurs. 10 and 13 Feb. 2025**)

- Omics in biomedical research
- Evidence and analytics
- Global Burden of Disease, Health Analytics, and Access to Therapeutics
- Economic Evaluations





Session 3

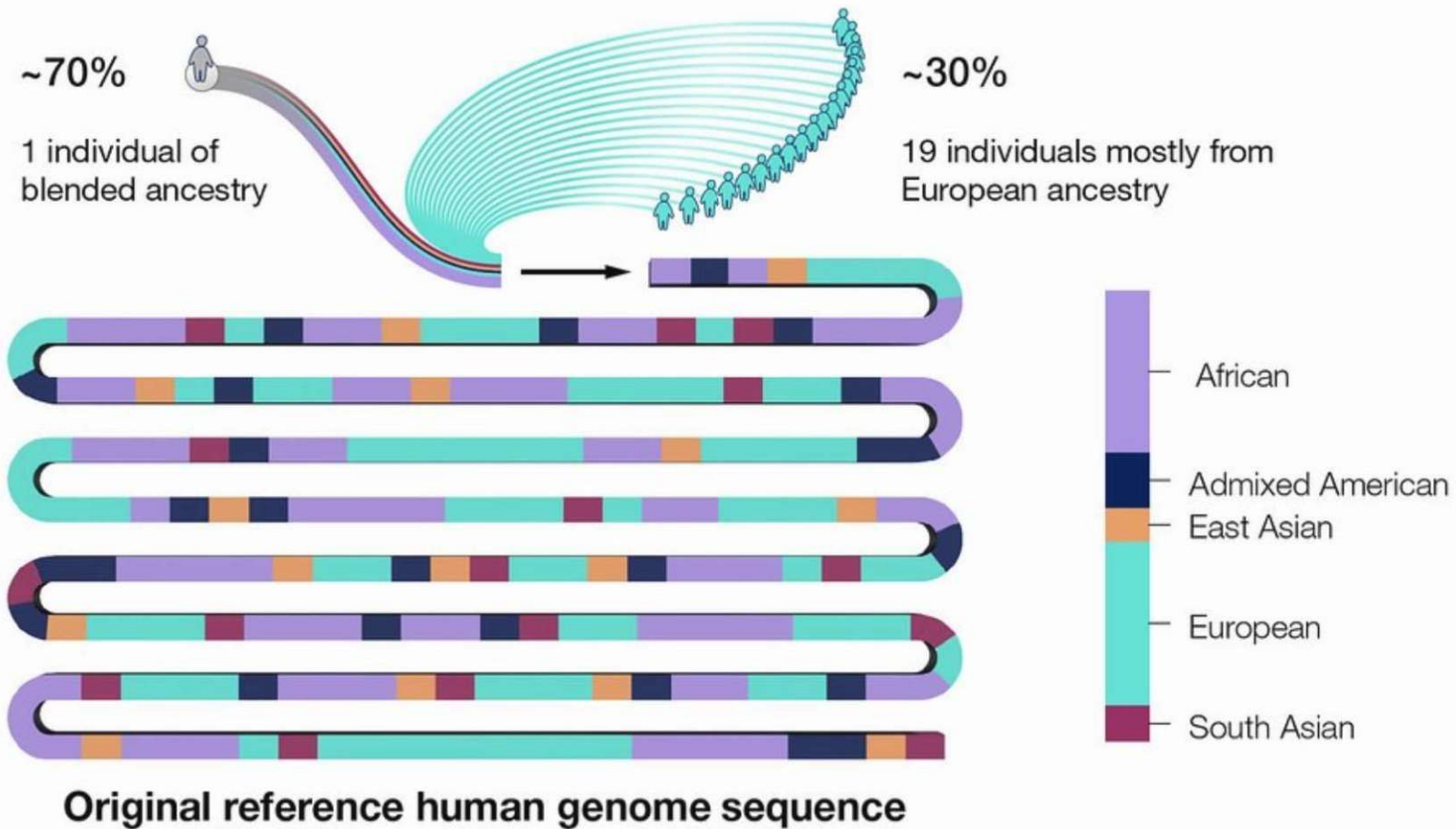
- Omics in biomedical research – **AI-based dry labs.**
- Evidence pyramid and analytics – **clinical vs. AI/omics-based studies**

From algorithm to bedside

Omics
pipelines

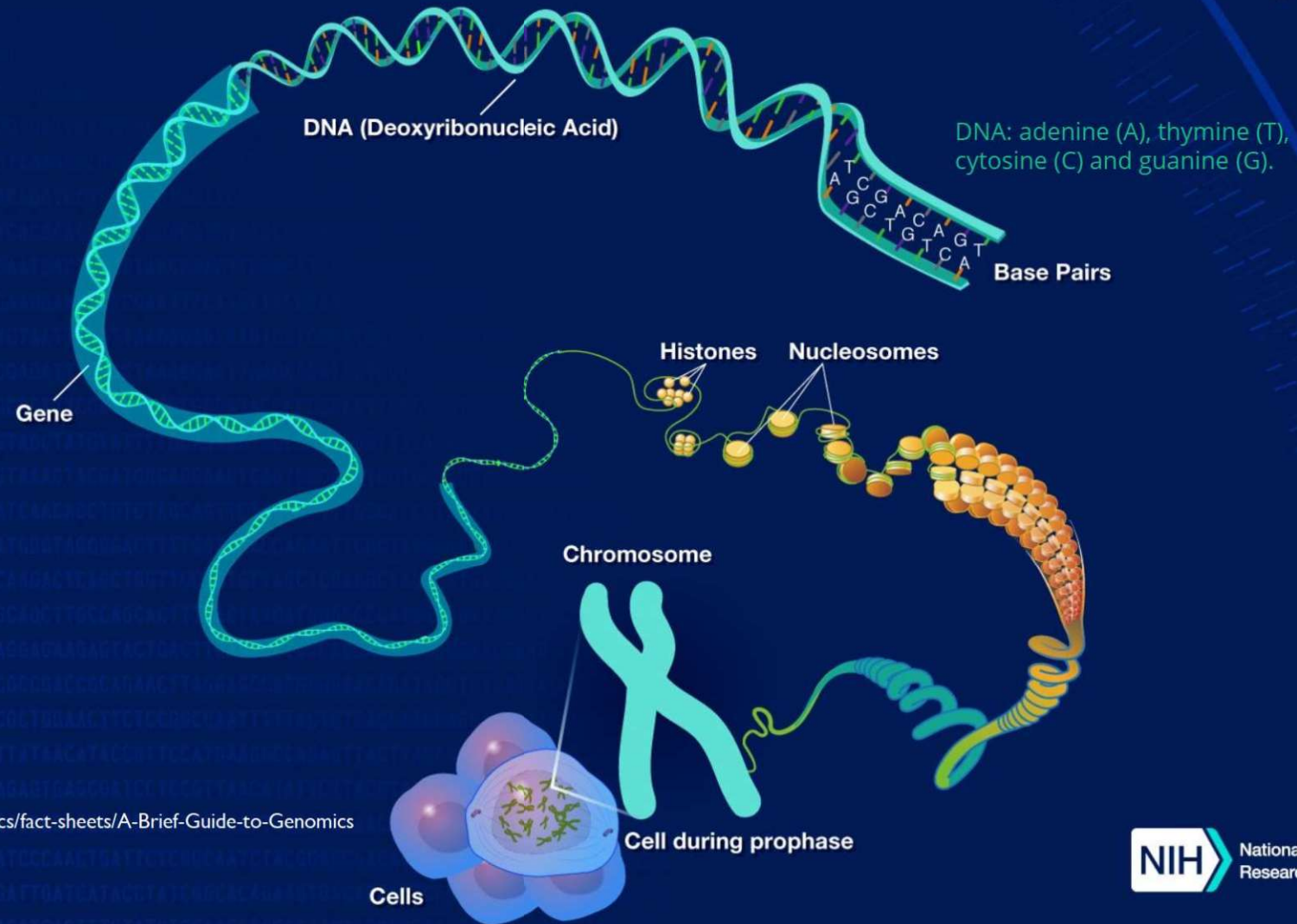


Human genome and diseases

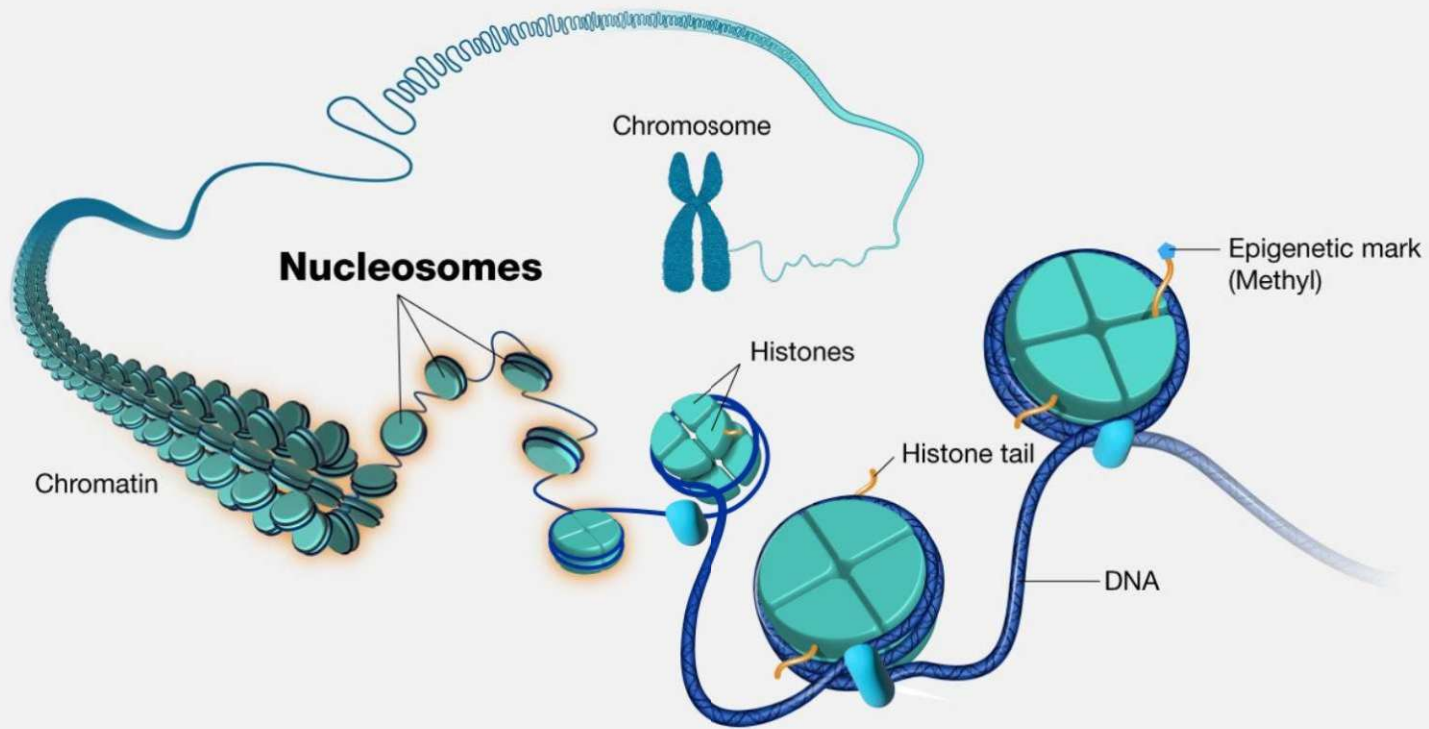


A Brief Guide to Genomics

NHGRI FACT SHEETS
genome.gov

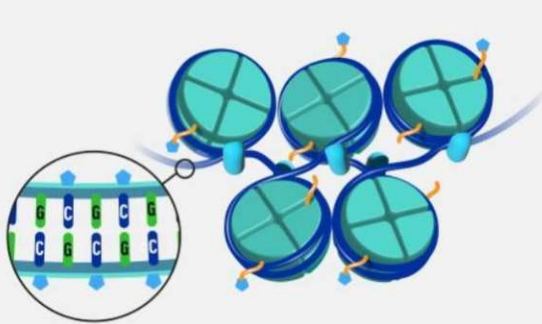


Source: <https://www.genome.gov/about-genomics/fact-sheets/A-Brief-Guide-to-Genomics>



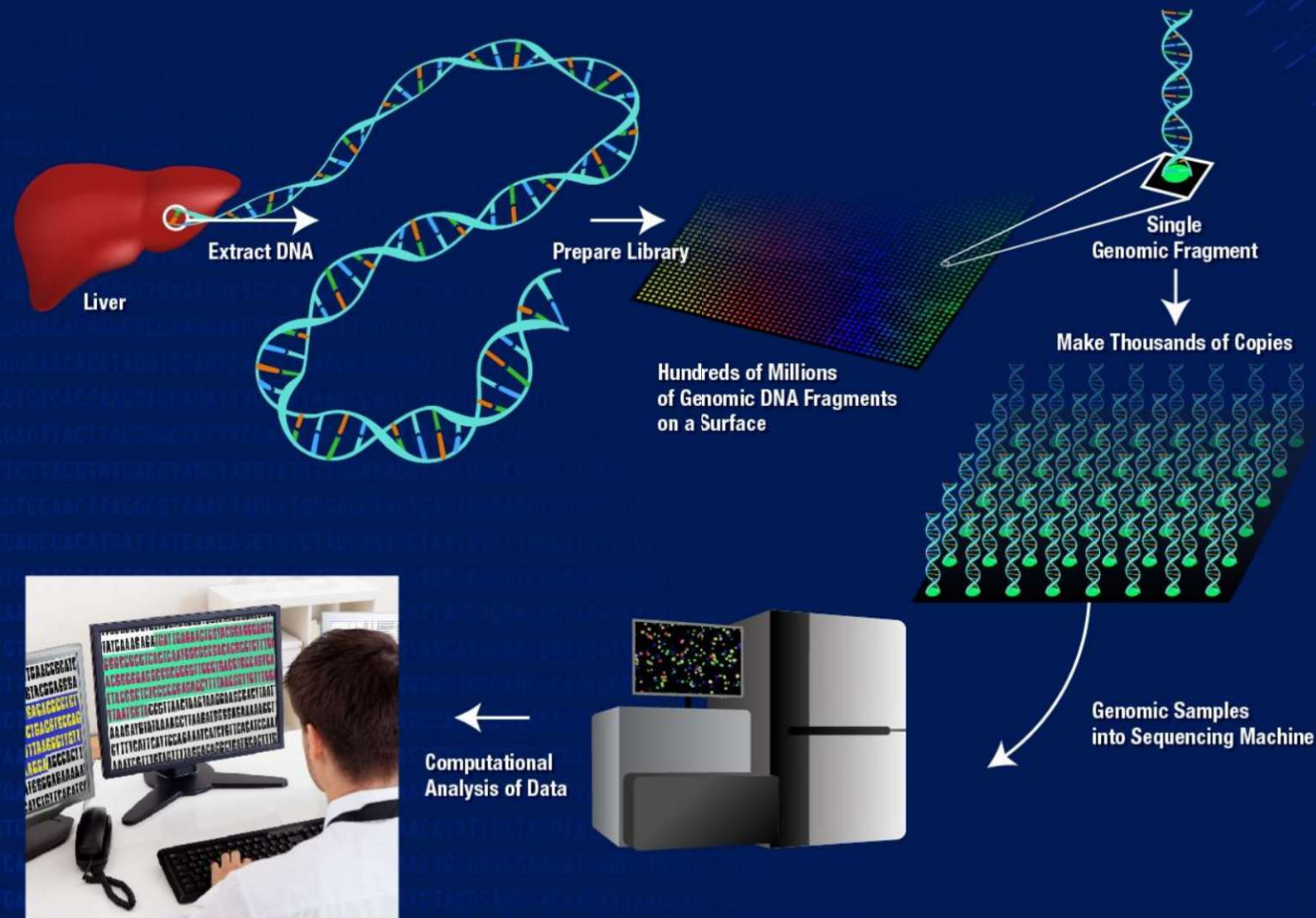
Closed chromatin (heterochromatin) is densely packed, and transcription cannot occur.

Open chromatin (euchromatin) is loosely packed, and transcription can occur.



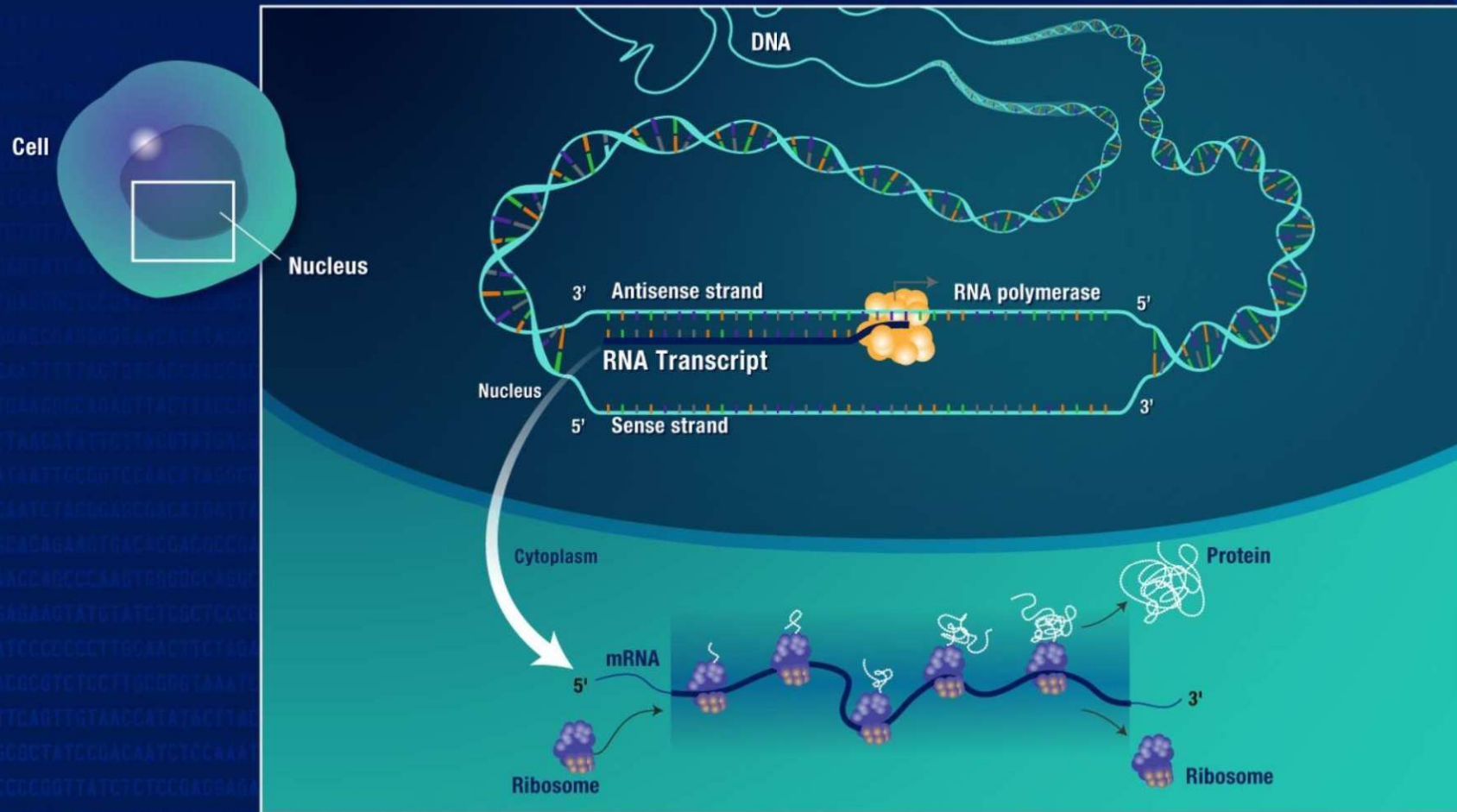
DNA Sequencing

NHGRI FACT SHEETS
genome.gov

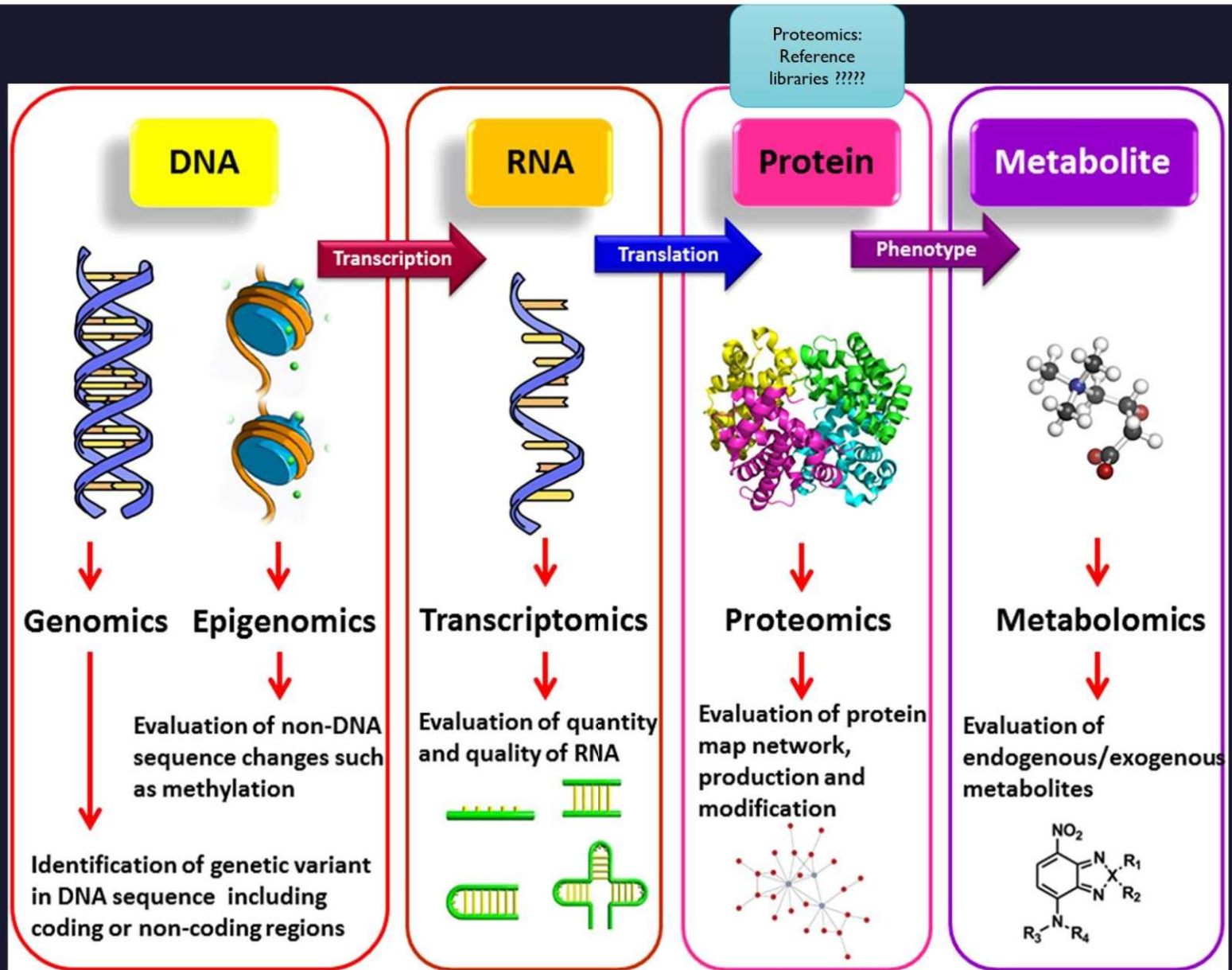


Transcriptome

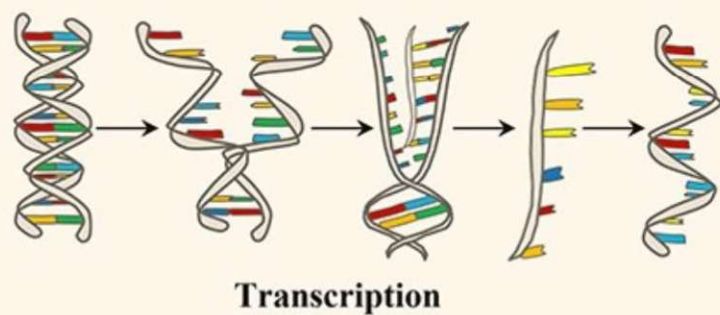
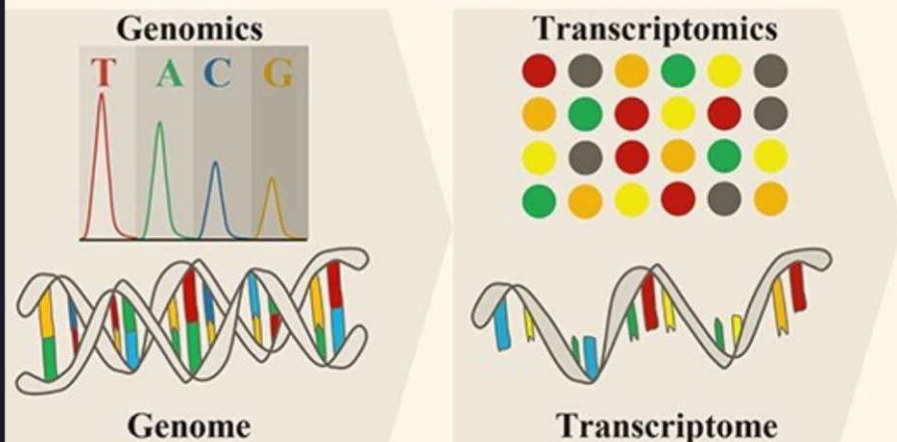
NHGRI FACT SHEETS
genome.gov



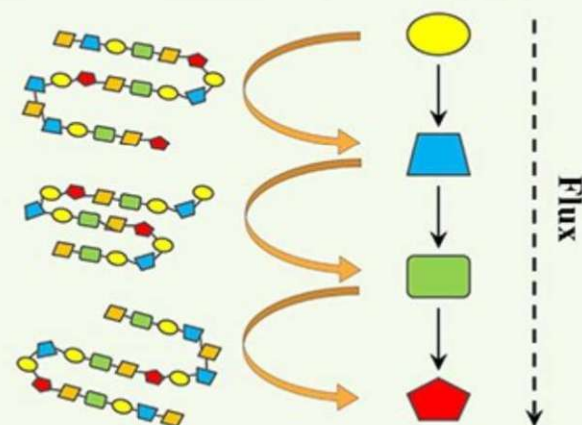
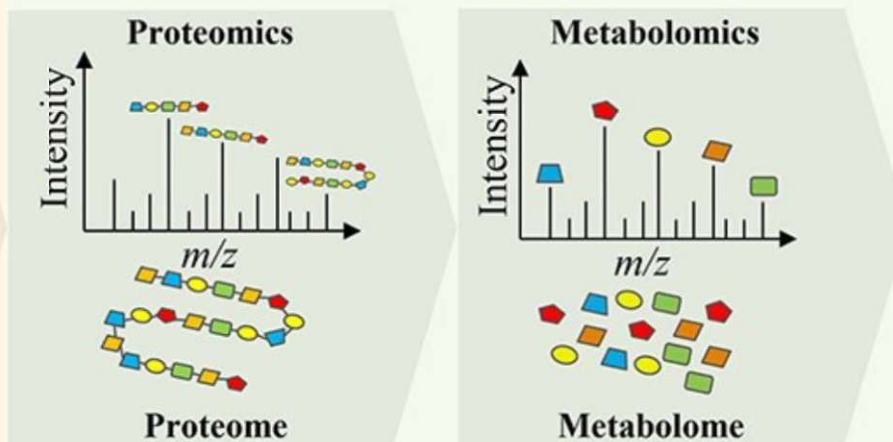
<https://www.genome.gov/about-genomics/fact-sheets/Transcriptome-Fact-Sheet>



Genotype



Phenotype



Function

<https://doi.org/10.3389/fbioe.2021.613307>



Omic

NIH definitions



28-01-2025

@NerminGhith

Genomics - The genome is the complete sequence of DNA in a **cell or organism**.

Transcriptomics - The transcriptome is the complete set of RNA transcripts from DNA in a **cell or tissue**.

Proteomics - The proteome is the complete set of proteins expressed by a **cell, tissue, or organism**.

Epigenomics - The epigenome consists of **reversible** chemical modifications to the DNA, or to the histones that bind DNA, and produce changes in the **expression of genes** without altering their base sequence.

Metabolomics - The metabolome is the complete set of small molecule metabolites found within a biological sample (including metabolic intermediates in **carbohydrate, lipid, amino acid, nucleic acid**, and other **biochemical** pathways, along with **hormones** and other **signaling** molecules, as well as exogenous substances such as **drugs** and their **metabolites**).

<https://www.ncbi.nlm.nih.gov/books/NBK202165/>



Genetic mapping

NHGRI FACT SHEETS
genome.gov

Genetic map



Cytogenetic map



Physical map

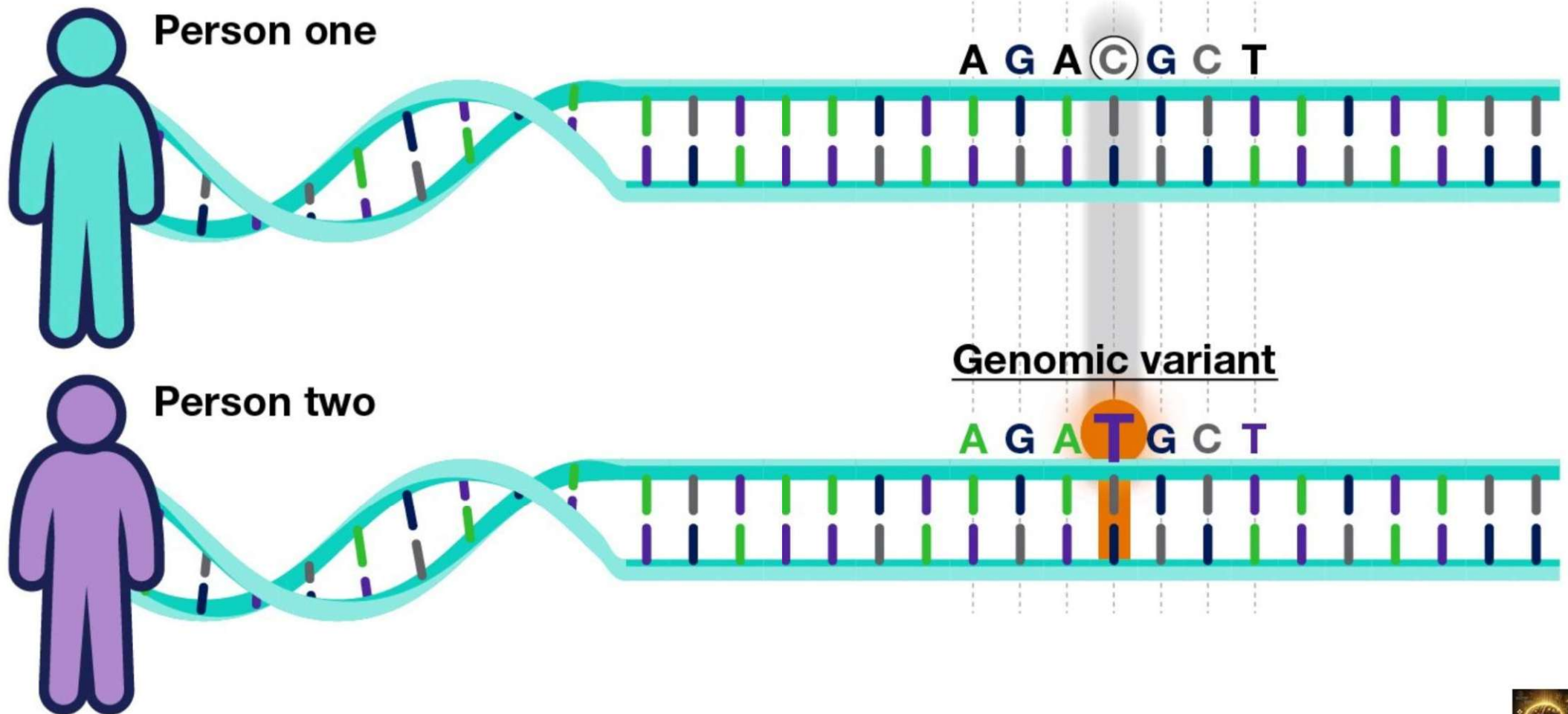


DNA sequence

....GATCTGCATGCATGCTAGCTAGCTAGCTAGCTAGAGCTTCG.... Bases

<https://www.genome.gov/about-genomics/fact-sheets/Genetic-Mapping-Fact-Sheet>

genomic variants and the risk for specific diseases

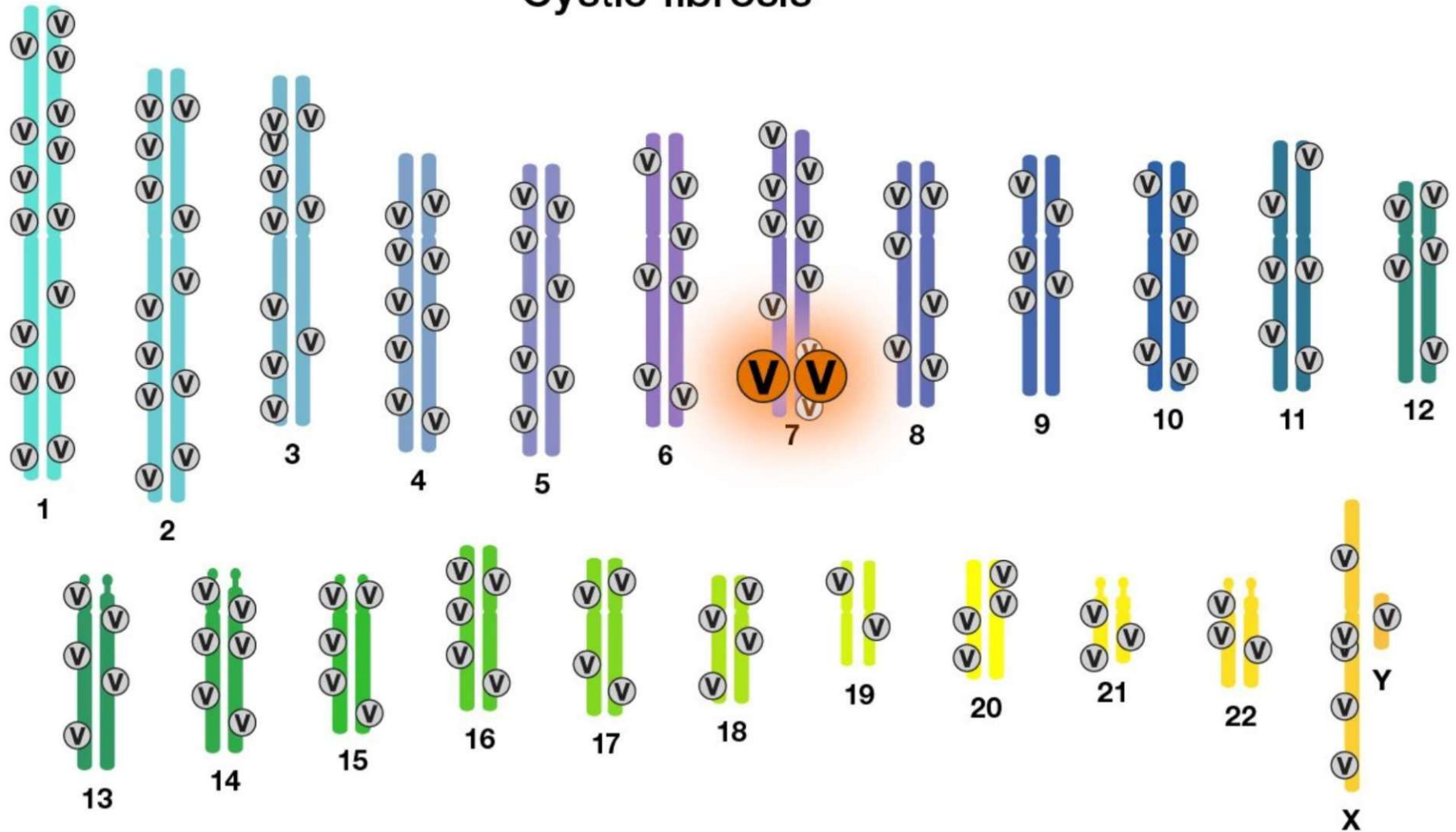


<https://www.genome.gov/Health/Genomics-and-Medicine/Polygenic-risk-scores>

Single-gene diseases

Cystic fibrosis

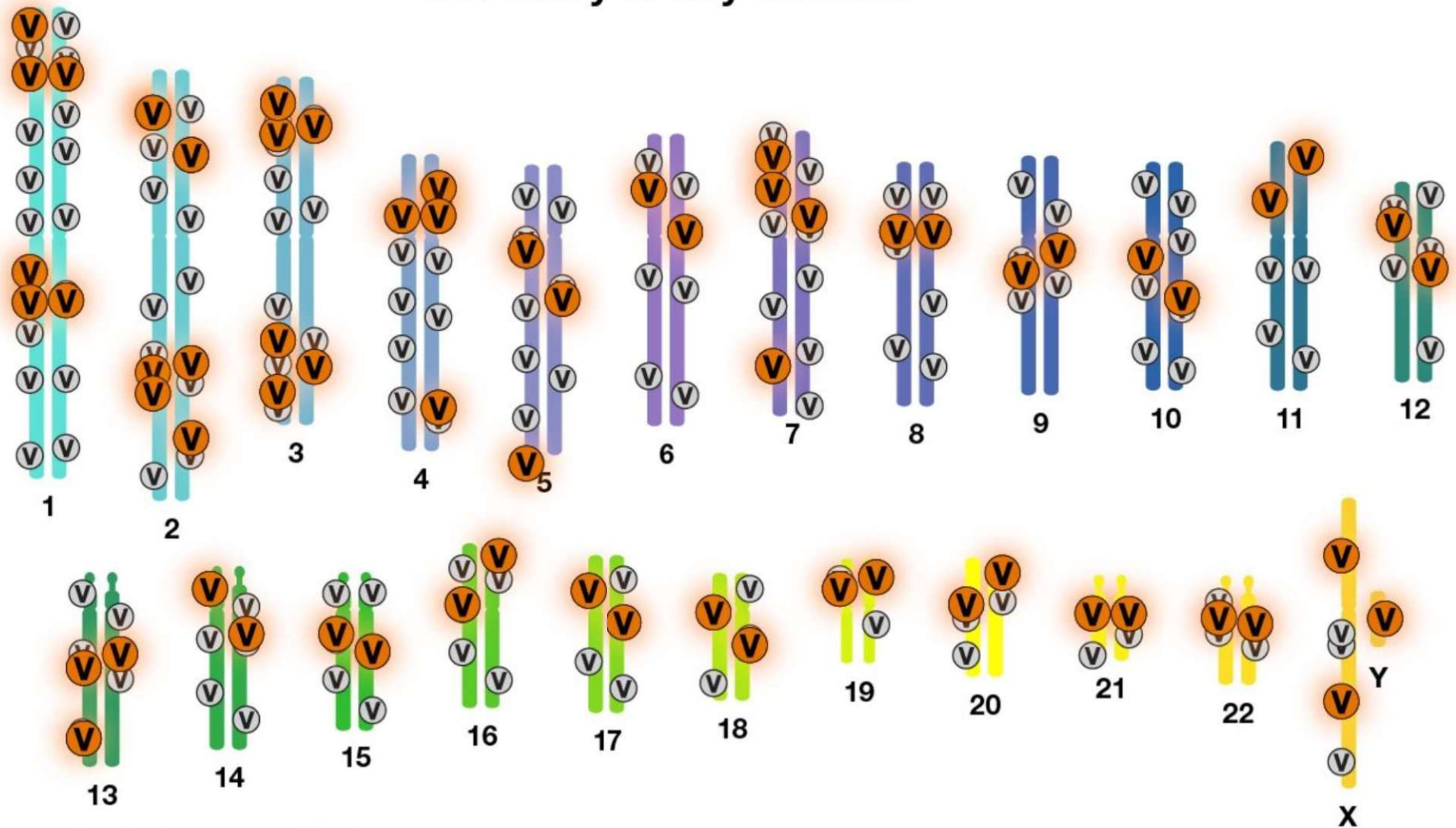
Person
one



Complex diseases - polygenic

Coronary artery disease

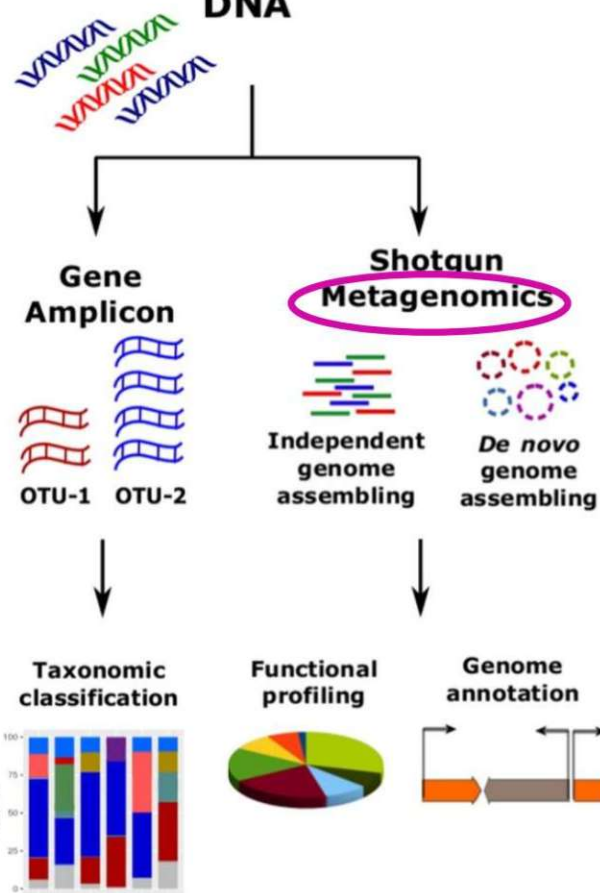
Person
two



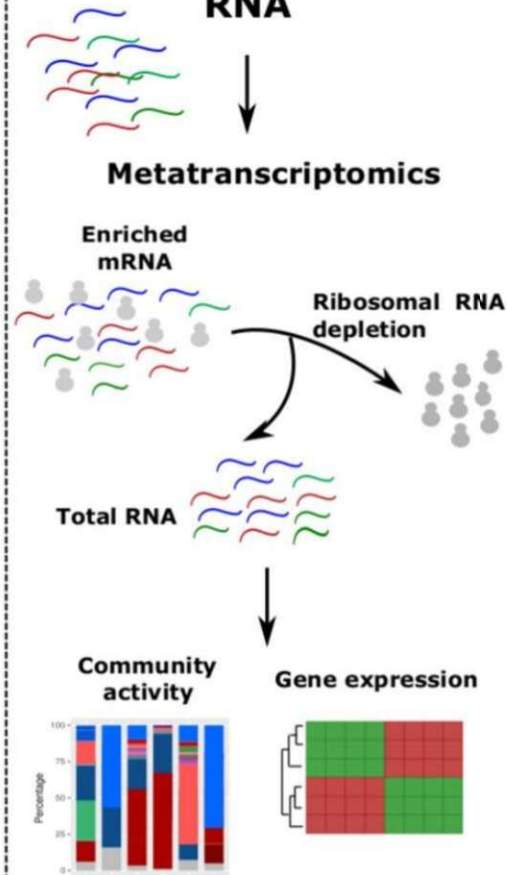
Microbial communities

Reference libraries ?????

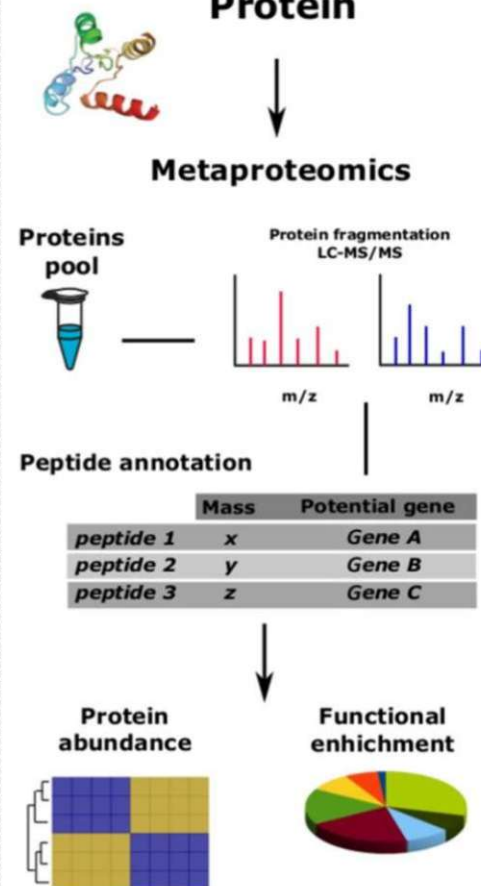
DNA



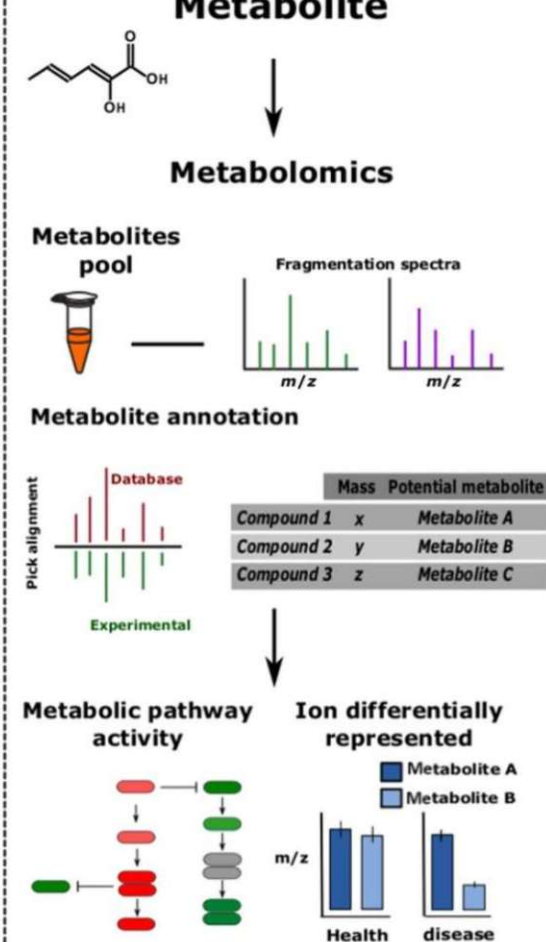
RNA



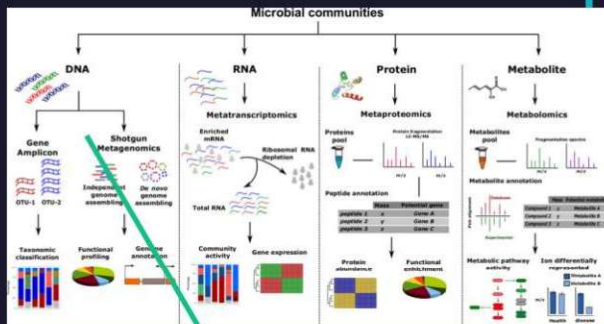
Protein



Metabolite



Own Project Pipeline



Fragments of DNA

19 Classes of Antibiotics

65 Phenotypes of Antibiotics

phenotypic resistance to different classes of antibiotics

Quantify per sample (i.e., FPKM)

Resistance Genes